

Name: _____

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Score: _____

Directions: Put away phones, calculators, smart watches, etc. Answer in the space provided. Show work.

1. For this problem, $A = \begin{bmatrix} 2 & 3 & -1 \\ 1 & 5 & 5 \end{bmatrix}$, $B = \begin{bmatrix} 2 & 1 \\ -4 & 3 \end{bmatrix}$, $C = \begin{bmatrix} -2 & 4 \end{bmatrix}$, and $D = \begin{bmatrix} -2 & 0 \end{bmatrix}$.

Perform the indicated operations or state that they are not possible.

(a) $BA =$

(b) $CBC^T =$

(c) $B^{-1} =$

(d) Find X if $C^T D - 2X^T = I_2$.

2. Suppose A , B and C are 3×3 matrices, and $\det(A) = 5$, and $\det(B) = 3$.

(a) $\det(A^2 B) =$

(b) $\det(2B^{-1}) =$

(c) Suppose $\det(ABC) = 10$. Find $\det(C)$.

3. Solve the system $\begin{cases} -3w + 6x - 3y - 9z = 12 \\ w - 2x + y + 3z = -4 \\ -\frac{1}{3}w + \frac{2}{3}x + \frac{1}{3}y - z = \frac{4}{3} \end{cases}$

4. Find x such that the matrix $\begin{bmatrix} 3 & x \\ -2 & -3 \end{bmatrix}$ is equal to its own inverse.

5. Find the determinant of the matrix $A = \begin{bmatrix} 3 & 5 & 2 & 4 \\ 1 & 2 & 5 & 2 \\ 0 & 1 & 2 & 3 \\ 0 & 0 & 0 & -1 \end{bmatrix}$.